| Student: | Email: |
|---|---|
| jeffar alaboudi | jmalab01@louisville.edu |

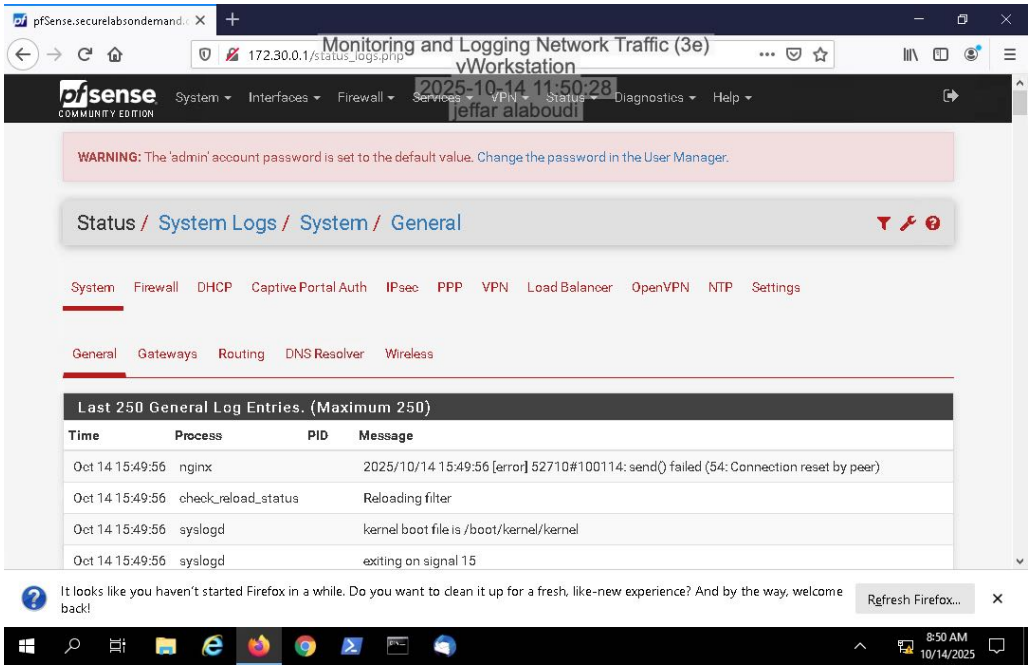
| Time on Task: | Progress: |
|---|---|
| 4 hours, 43 minutes | 100% |

Report Generated: Sunday, November 30, 2025 at 6:28 PM

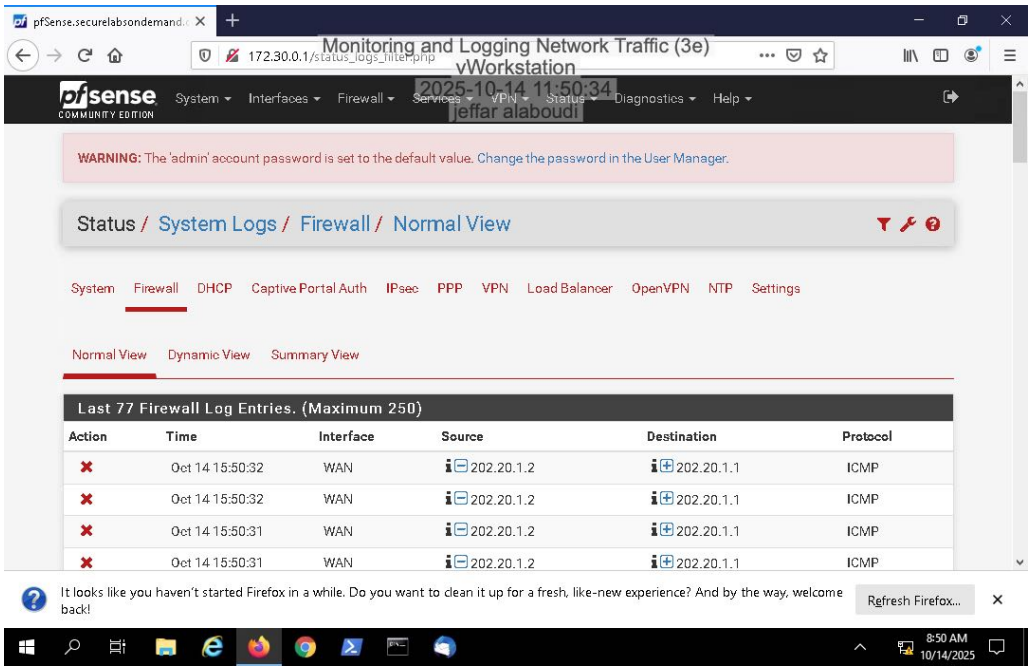
# Section 1: Hands-On Demonstration

## Part 1: Configure the pfSense Firewall Log

13. **Make a screen capture** showing the **system logs**.

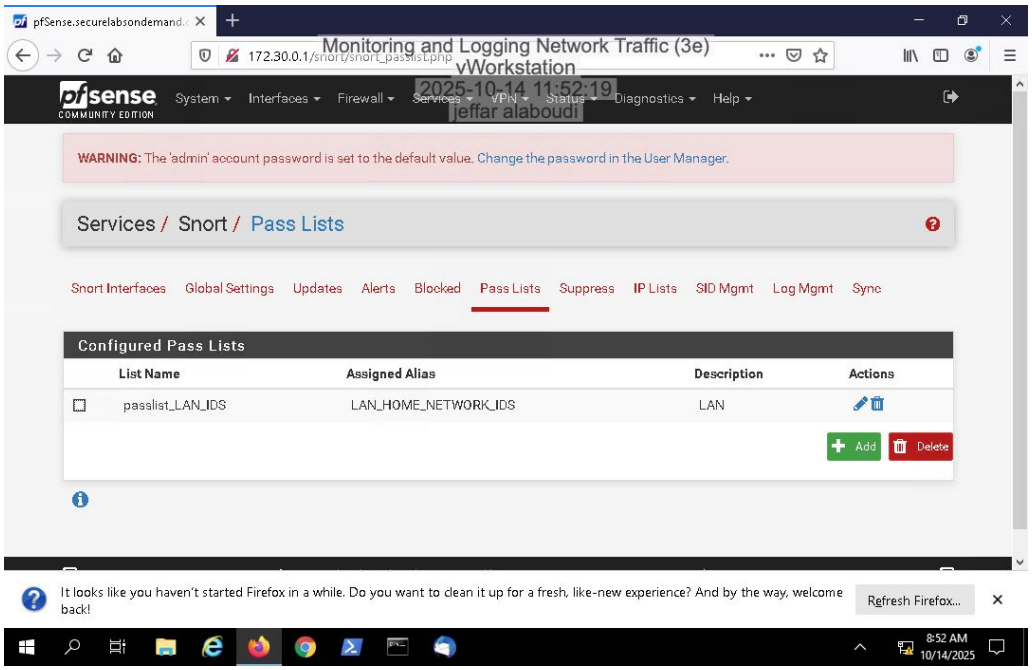


15. **Make a screen capture** showing the **firewall logs**.

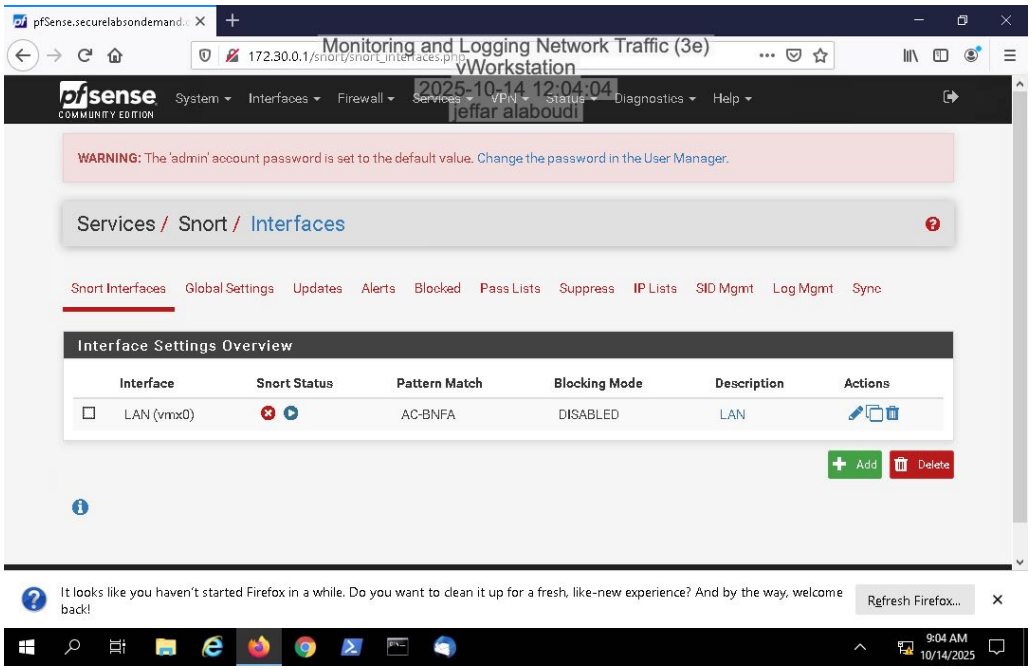



## Part 2: Configure a Snort Intrusion Detection System

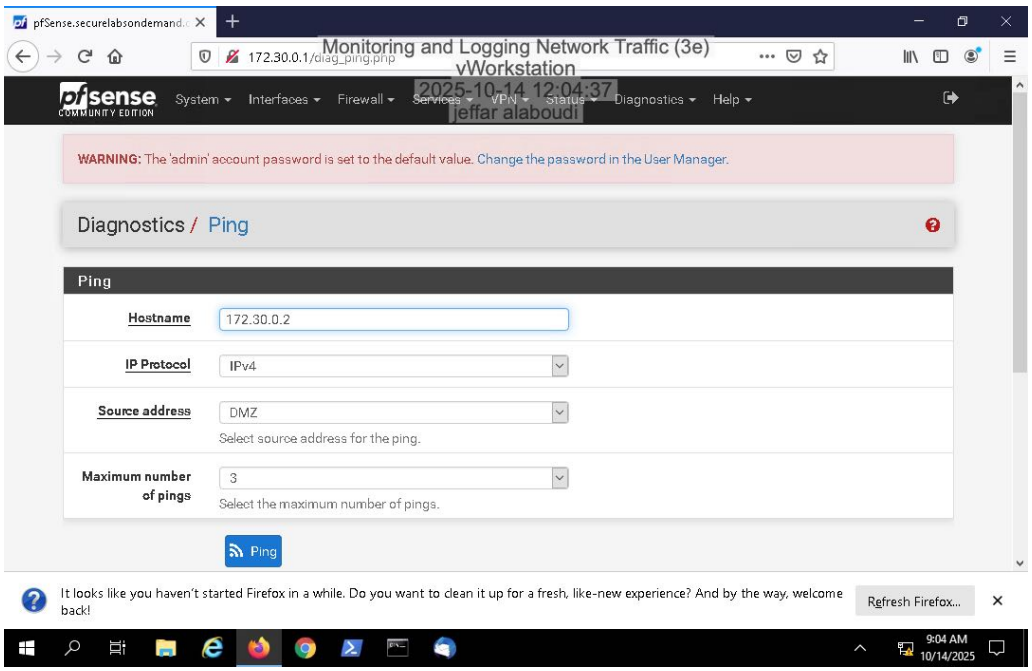
14. **Make a screen capture** showing the **updated Pass Lists page**.



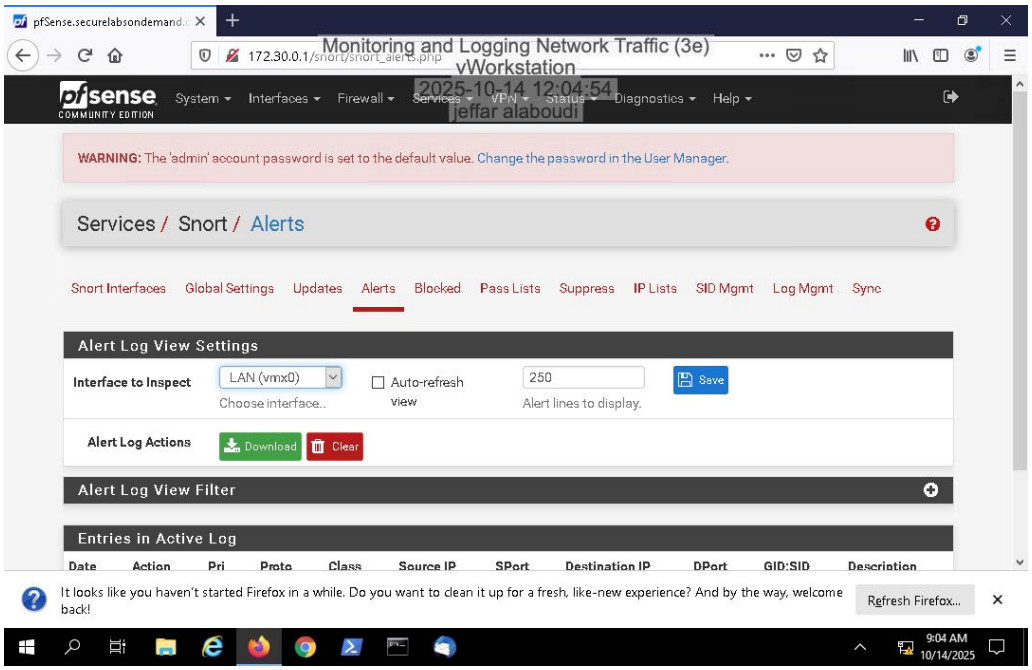
28. **Make a screen capture** showing the **active Snort status on the LAN interface**.




33. **Make a screen capture** showing the **successful ping results**.

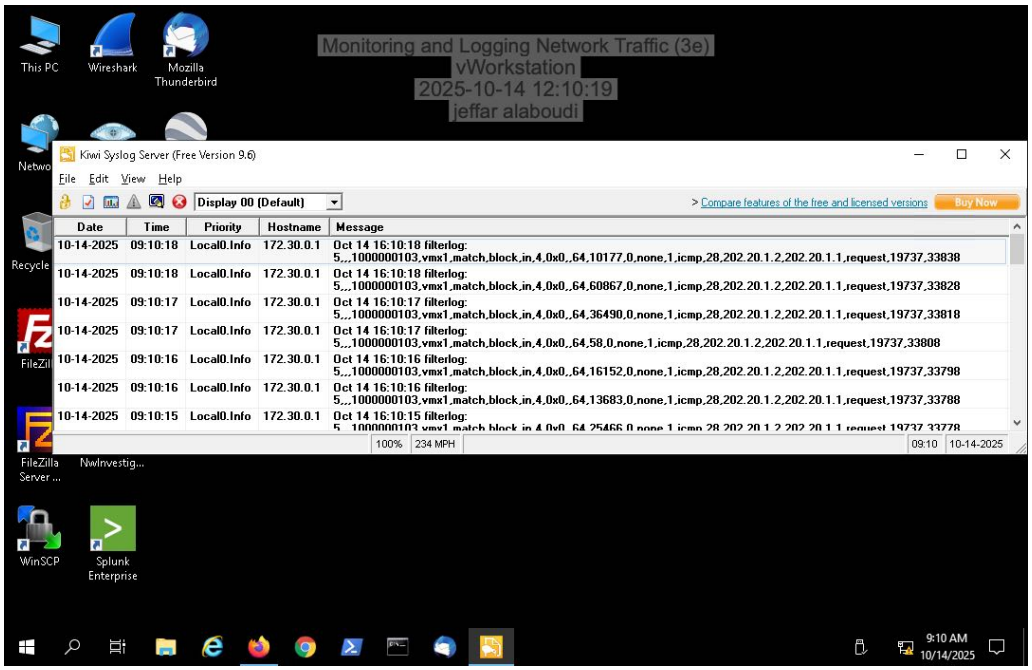


38. **Make a screen capture** showing the **ICMP alerts in the Snort Active Log**.




## Part 3: Implement Firewall Log Forwarding with Kiwi Syslog Server

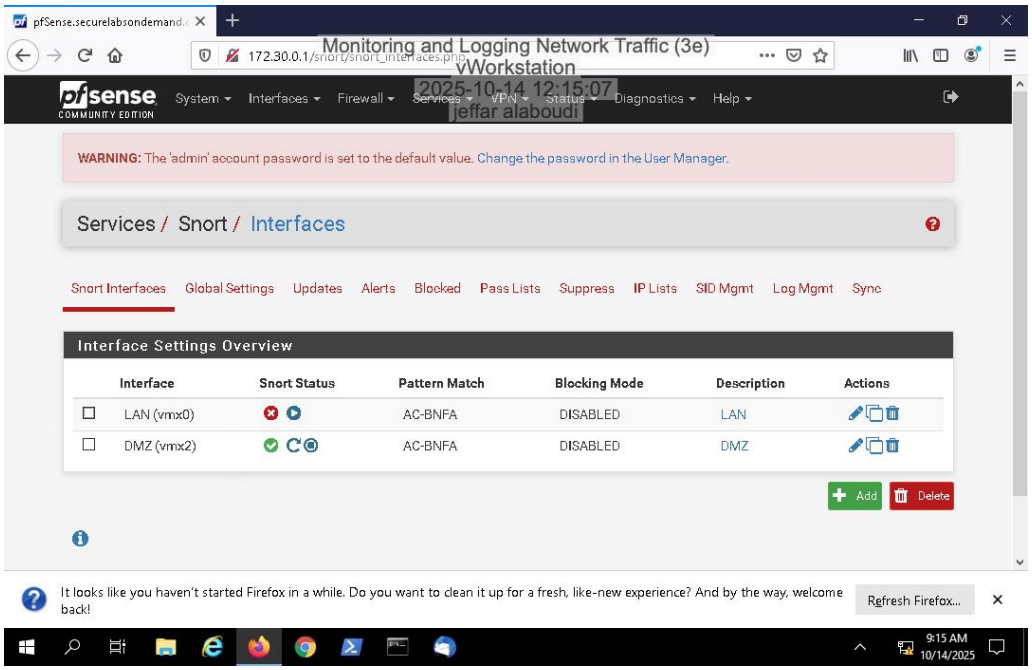
17. **Make a screen capture** showing the **pfSense firewall log events in Kiwi Syslog Server**.



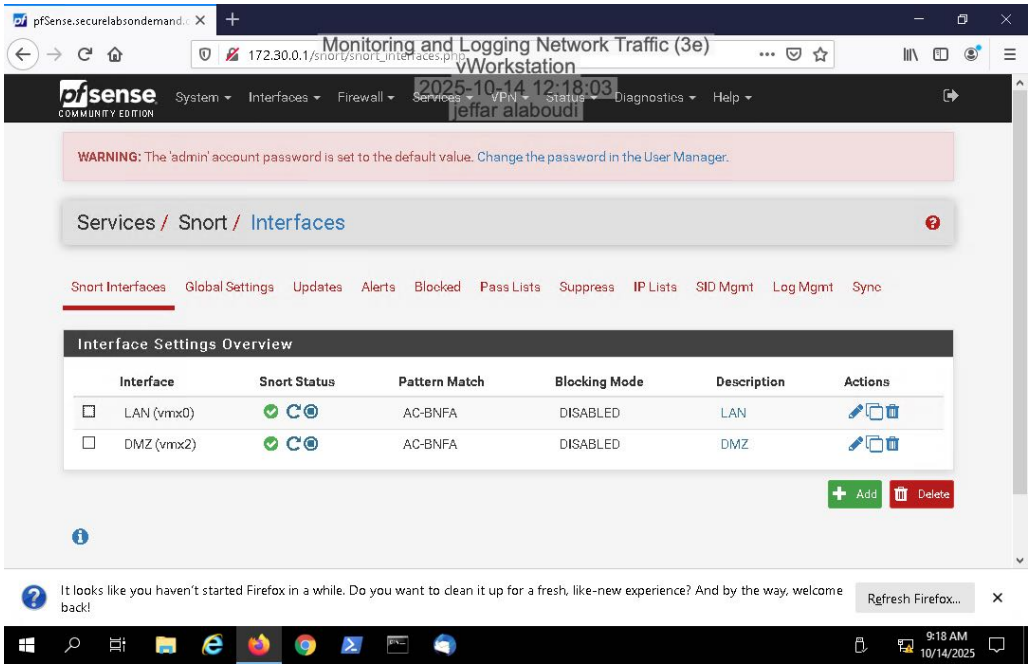
# Section 2: Applied Learning

## Part 1: Configure Snort Monitoring on the DMZ

17. **Make a screen capture** showing the **active Snort status on the DMZ interface**.

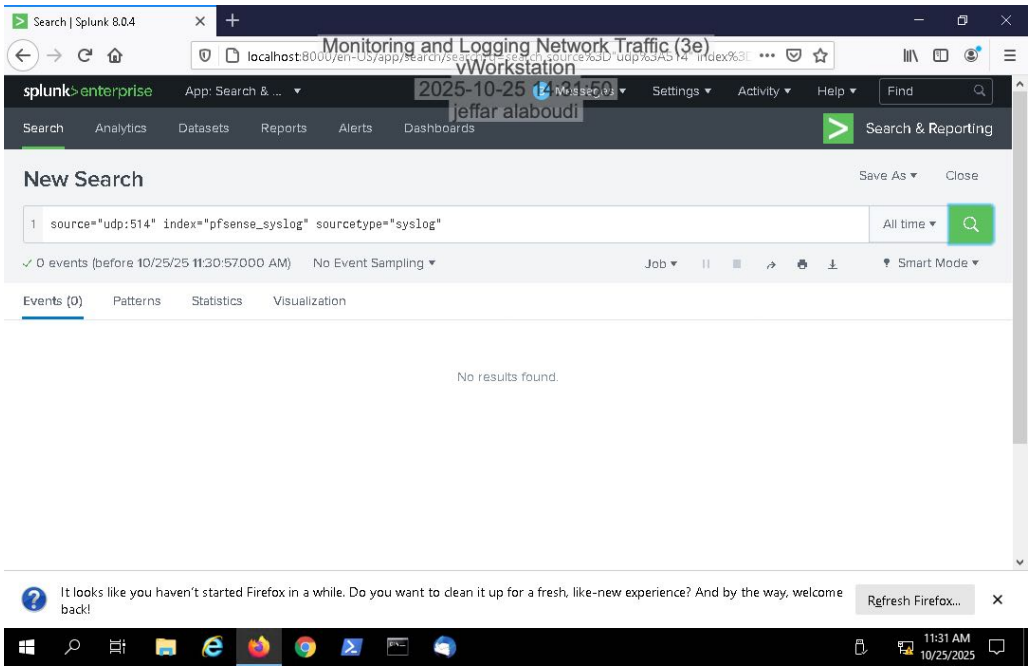



20. **Make a screen capture** showing the **Snort GPLv2 Community Rules enabled and "live-reloading" message**.



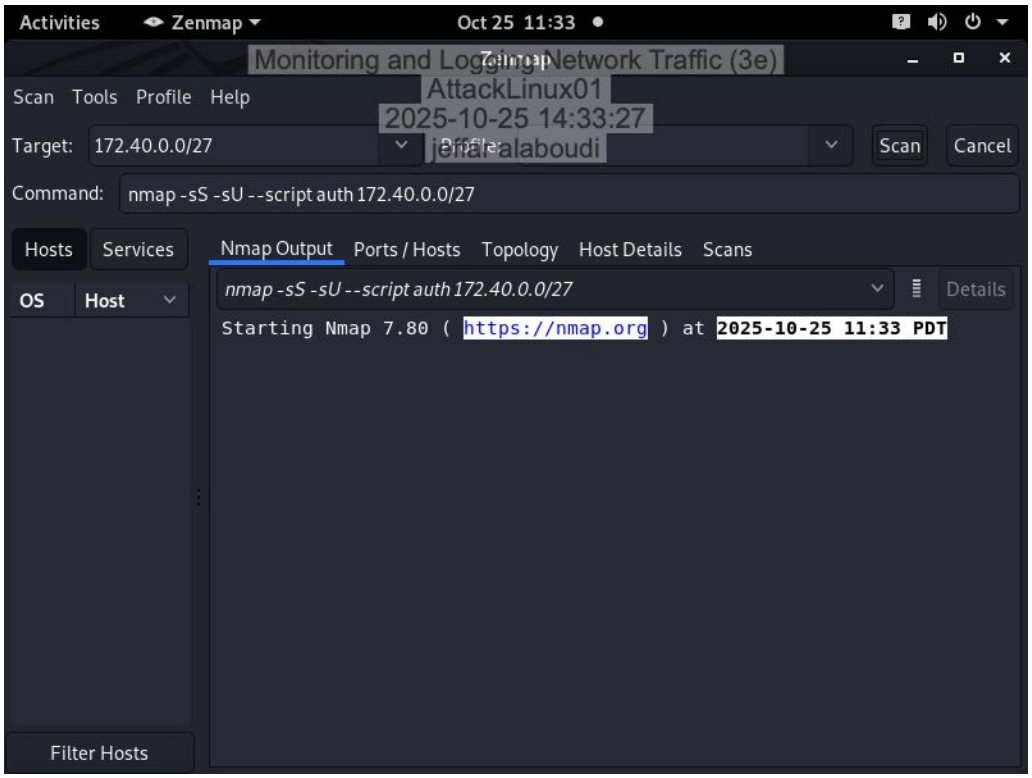
## Part 2: Implement Security Information and Event Management with Splunk

13. **Make a screen capture** showing the **indexed events in Splunk**.

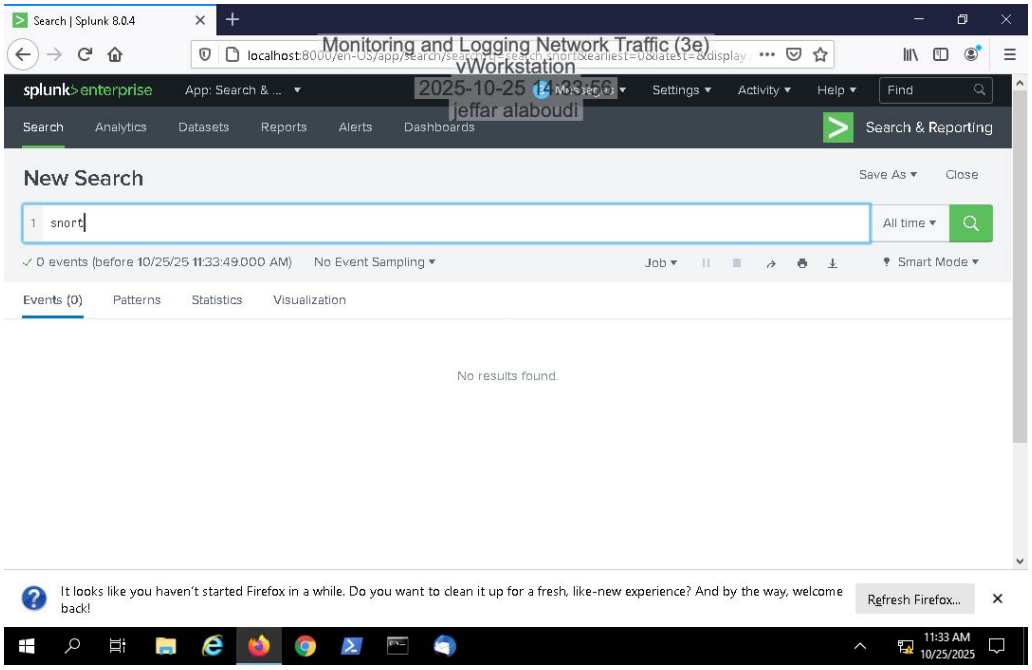



## Part 3: Simulate and Detect a Perimeter Network Attack

6. **Make a screen capture** showing the **Nmap scan report**.



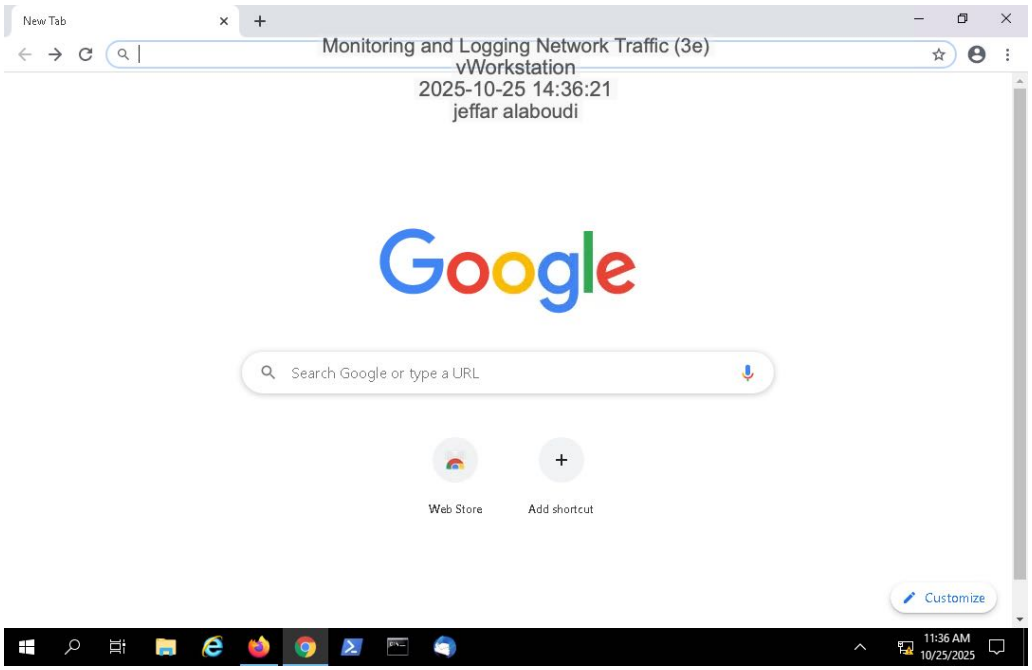

9. **Make a screen capture** showing the **search results in Splunk**.



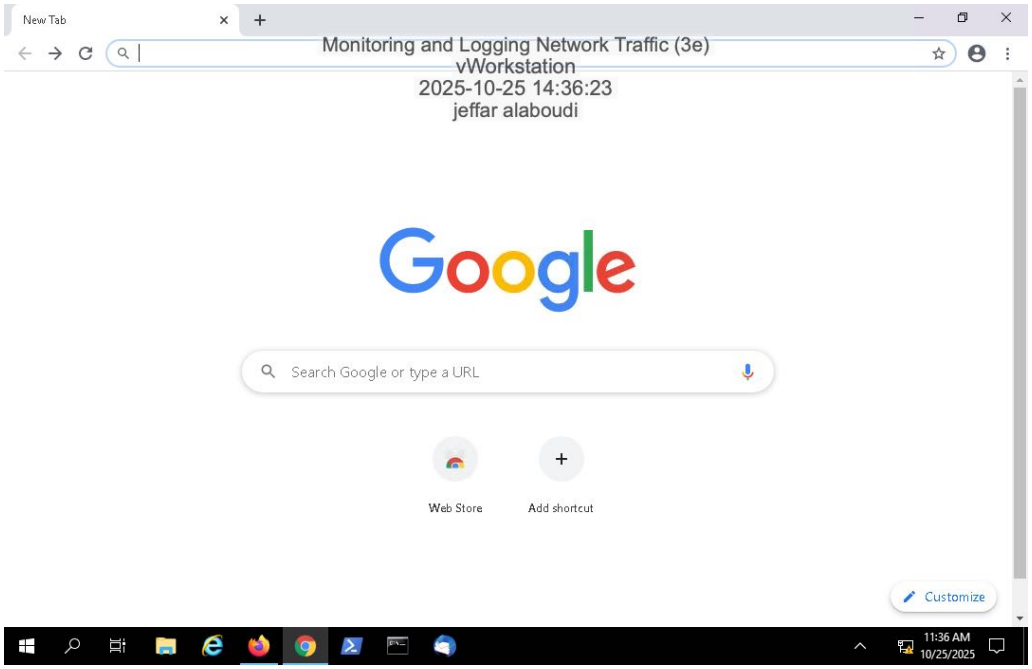
# Section 3: Challenge and Analysis

## Part 1: Simulate a DMZ Breach with Infection Monkey

**Make a screen capture** showing the **resulting Infection Map**.




**Make a screen capture** showing the **resulting Security Report**.

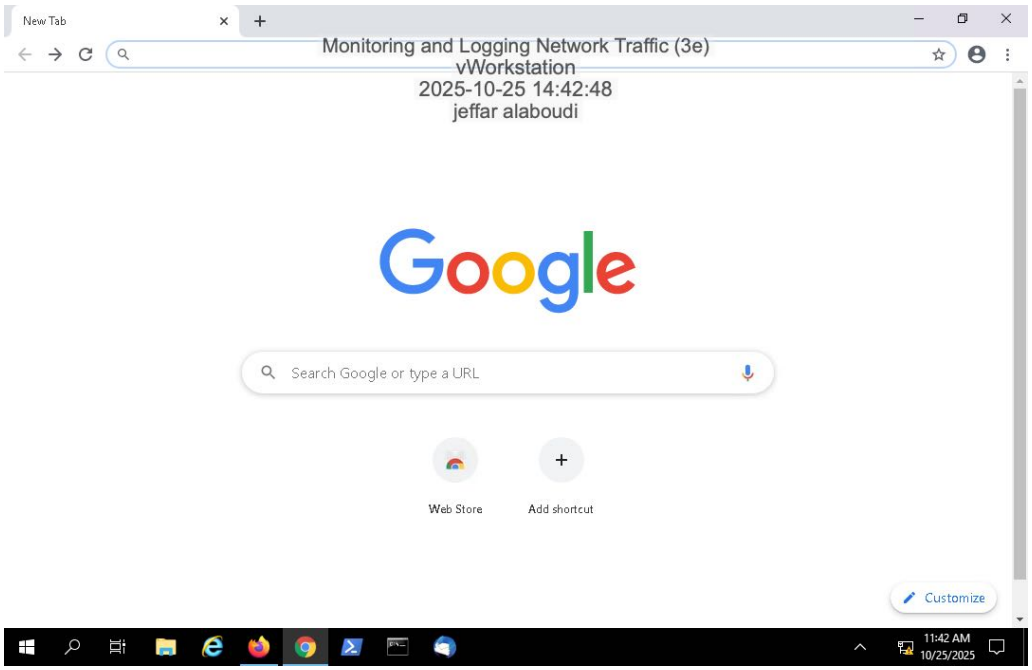


**Summarize** your DMZ breach simulation results, highlighting what you found to be the greatest concerns from a network monitoring perspective.

during the dmx breach simulation one of the greatest concerns identified was the lack of timely detection of abnormal traffic patterns the monitoring tools shows spikes in inbound and outbound traffic that initially blended with legitimate activity making it difficult

## Part 2: Detect a Simulated DMZ Breach with Snort and Splunk

**Make a screen capture** showing the **results of your search query for Infection Monkey traffic in Splunk**.




**Describe** any concerns about the structure of the query result or the data elements it contains. What data fields would you add, remove, or edit to make log analysis more effective?

one concern with the structure of the query results is that many logs consistent formatting and may omit important context which makes correlation across different systems more difficult for example some entries may contain only source ip addresses without destination ports, users id, or session details , limiting the ability to trace activity back to a specific host or account.

**Write a brief memo** to your manager describing Splunk's usefulness in detecting traces of your simulated breach. What configuration changes would you recommend? How would you enhance its functionality?

During the simulated breach exercise, Splunk proved highly valuable in detecting suspicious activity by aggregating and correlating logs from multiple sources. Its search and alerting capabilities allowed early identification of abnormal traffic patterns, unauthorized login attempts, and potential lateral movement across the network. Splunk's real-time dashboard also provided visibility into the timeline of events, which is critical in reducing response time.